



Quant Week 1

1/30/25



Agenda



1. Icebreaker Question
2. Brainteasers
3. Order Book
4. Market Making
5. High Frequency Trading
6. Project



Ice Breaker

- Tell me what is your favorite movie and why.





Brain Teaser - Glass Balls



You are holding two glass balls in a 100-story building

If a ball is thrown out of the window, there is a certain floor X where every throw at or above that floor will break the ball, but the ball will not break if thrown under that.

A ball can be rethrown if it does not break.

At most how many throws will it take to determine X ?



Brain Teaser - Answer

Let's say the maximum amount of **throws** for a ball is N .

You would have to drop the first ball on the N th floor.

If that breaks you try floors 1 to $N-1$ in order

If that does not break you then drop the same ball again on the floor $N + N-1$

You then have $N-2$ throws with the other ball to try floors $N+1$ to $N-2$

For N throws you would be able to cover $N + N-1 + N-2 + \dots + 1 = N(N-1)/2$ (summation formula)

In this case $N(N+1)/2 \geq 100$, so $N \geq 14$.

It takes at most 14 drops to find floor X .

What is an Order Book?





Order Book



- A visual representation of current limit orders
 - Limit orders: Orders placed to buy or sell at a specific, predetermined price
 - Market orders: Orders placed to buy at the current available price
- Understanding an order book is very important for working in quant (and as a result, interviewing for quant)
- The order book is a key component of understanding how market makers make money

Order Book Example



Bid (Volume)	Price	Ask (Volume)
	100.03	150
	100.02	100
	100.01	50
	100	
60	99.99	
120	99.98	
180	99.97	

Order Book Example



Bid (Volume)	Price	Ask (Volume)
	100.03	150
	100.02	100 (SOLD)
	100.01	50 (SOLD)
	100	
60	99.99	
120	99.98	
180	99.97	

Order Book Example



Bid (Volume)	Price	Ask (Volume)
	100.05	250
	100.04	200
	100.03	150
	100.02	
60	100.01	
120	100.00	
180	99.99	

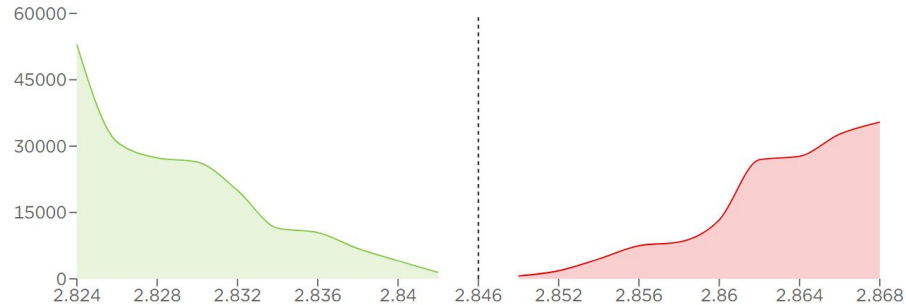
Order Book Example



Order book

Order	Volume	Buy	Sell	Volume	Order
2	1452	2,842	2,850	647	2
4	2635	2,840	2,852	1179	1
2	2756	2,838	2,854	2656	4
3	3621	2,836	2,856	3009	2
2	979	2,834	2,858	829	3
3	8566	2,832	2,860	5000	1
4	6382	2,830	2,862	13587	3
2	958	2,828	2,864	804	2
2	3596	2,826	2,866	5000	1
5	22030	2,824	2,868	2714	2

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Market Making





What is Market Making?



In the general sense, a market is a place where you can both buy and sell assets (food, clothing, financial assets, illegal goods, etc.).

Any person who buys or sells a given asset might be a component of a market, but is not considered a market maker. The key feature of a market maker is that they quote both a buy **and** sell price for an asset.

Some of the more well known market makers are:

- Citadel Securities
- Jane Street
- Virtu Financial
- Many large banks (Goldman Sachs, Morgan Stanley)



How do MMs make money?



The goal of a market maker is to quote bid and ask prices such that people buy and sell equal amounts of the asset on both sides.

From the order book example earlier, if there are no large orders, and market orders are trickling in, whoever is making that market makes \$0.02 for every pair of offsetting trades.

An interesting feature of this strategy is that market makers remain neutral to moves up or down in the market. As long as their quoted prices result in equal amounts of trading on either side, they profit, regardless of large swings in the market.



How do MMs make money?



Market makers stay in the middle of the market by continuously adjusting their bid and ask prices to reflect changing conditions. If prices rise, they increase both bid and ask prices accordingly, and if prices fall, they lower both. This ensures that they always remain competitive and can profit from the bid-ask spread. Additionally, they manage their inventory carefully. If they accumulate too much of an asset due to falling prices, they lower their bid to slow down buying and adjust their ask to sell off inventory. Conversely, if they sell too much during rising prices, they raise their bid to replenish their holdings while also increasing their ask price.

Another key strategy is order flow prediction. Market makers analyze order book imbalances to anticipate future price movements. When buy orders dominate, they adjust their quotes upward to stay competitive, and when sell orders dominate, they adjust downward. Many modern market makers rely on high-frequency trading (HFT) algorithms to make these adjustments rapidly, ensuring they always maintain a competitive spread while minimizing inventory risk.

To further protect themselves, market makers hedge against large market moves. When volatility increases, they may use options, futures, or correlated assets to offset risk and maintain stability. Additionally, they provide liquidity at multiple price levels rather than placing a single bid and ask order. This allows them to scale in and out of positions smoothly, ensuring that they can continue facilitating trades without exposing themselves to excessive risk. By employing these strategies, market makers remain neutral in the market while consistently profiting from the spread, regardless of large price swings.

High Frequency Trading





What is HFT?



The practice of trading at incredibly high speeds

- Driven by highly efficient algorithms written in low level languages or on FPGAs
- Requires significant infrastructure and investment
- Generally profit per trade is incredibly low, but the sheer volume of trades results in large profits

HFTs are incredibly secretive about their strategies. As soon as another firm discovers their strategy, the pseudo-arbitrage opportunity closes, and it is generally no longer profitable.

As a result, many employees are required to sign NDAs and long non-competes (2+ years).

Basic HFT Strategies





Arbitrage

Arbitrage describes an opportunity in which an individual can buy an asset, and immediately sell it for a higher price.

The most common example is in ForEx markets, but it can also occur between exchanges.

Because of the competition present, these opportunities rarely exist, and are closed quickly when they do.



A decorative geometric shape in the top-left corner, consisting of a white triangle pointing right and a dark blue triangle pointing left, overlapping each other.

Latency Arbitrage



Latency Arbitrage is the most pure example of HFTs' advantage over other traders/investors.

- Suppose a hedge fund needs to buy a large chunk of a stock, and there isn't enough liquidity in the order book
- A HFT firm sees the hedge fund buying up all of this stock (in the form of rapid price movements) and starts to buy all of the stock on other exchanges
- As a result, the hedge fund now has to pay a higher price, and the difference is the HFT firm's profit



Event Arbitrage



Event arbitrage is when a HFT makes use of its speed to react to news stories faster than other players in the market

- With the Houthi attacks on shipping lanes, ships now have to sail around the Horn of Africa, instead of cutting through the Red Sea
- This results in significantly higher oil costs
- An HFT firm might see this and be able to react far more quickly than other players



Statistical Arbitrage

- Essentially a more complicated version of pairs trading
- Firms look at correlation between assets, and use highly technical models to bet on the convergence or divergence of certain assets or indicators
- Generally more common nowadays



Project Discussion/Work Time





Attendance Form

This one



[Link to Onboarding Document, Slack
and Team Folder](#)